RA8P1 E-READER

Development draft — not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

128 MiB Octal NOR



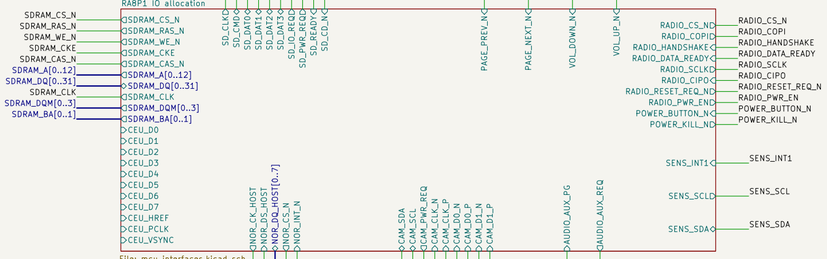
File: nor_flash.kicad_sch

Page and volume buttons

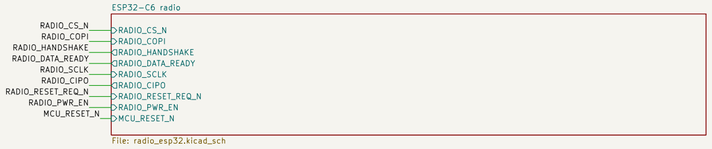


File: user_controls.kicad_sch

RA8P1 IO allocation

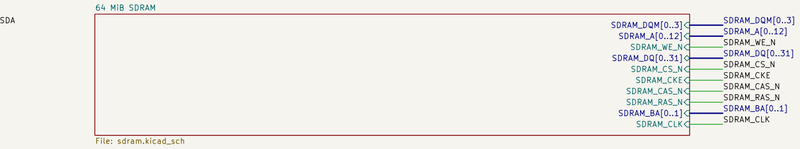


ESP32-C6 radio



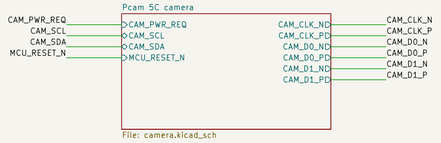
File: radio_esp32.kicad_sch

64 MiB SDRAM



File: sdram.kicad_sch

Pcam 5C camera



File: camera.kicad_sch

Power button and source control



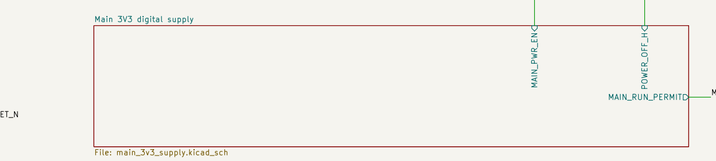
File: power_button.kicad_sch

Orientation sensor



File: orientation_sensor.kicad_sch

Main 3V3 digital supply



File: main_3v3_supply.kicad_sch

RA8P1 clocks, reset and debug



File: mcu_clocks_debug.kicad_sch

Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

microSD removable storage



File: microsd.kicad_sch

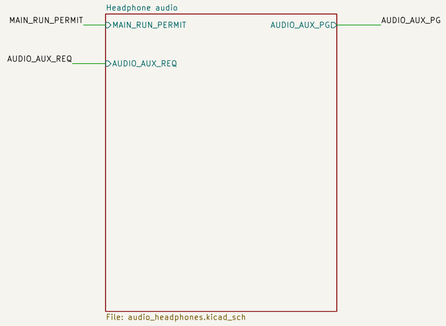
RA8P1 internal core regulator

File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies

File: mcu_supply_power.kicad_sch

Headphone audio



File: audio_headphones.kicad_sch

Sheet: /	
File: ereader_rev1.kicad_sch	
Title: RA8P1 e-reader — schematic index	
Size: A2	Date: 2026-09-05
KiCad E.D.A. 10.0.6	**Rev: 0.3 DRAFT**
	Id: 1/15

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

+3V3_MCU FB1 120R @100MHz +3V3_USBHS_A PWR_FLAG C43 10u GND

USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm: 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R: initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification: verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]
ERC: #FLG03 declares the +3V3_MCU feed through FB1.
This does not qualify the pending upstream SYS_AON source.

U1B R7KA8P1KFLCAC#UC0

GPIO PO / P1

U13	P000	P100	U6	NOR_DQ_HOST0
T13	P001	P101	R5	NOR_DQ_HOST3
U14	P002	P102	P5	NOR_DQ_HOST4
R14	P003	P103	R4	NOR_DQ_HOST2
T11	P004	P104	M6	NOR_CS_N
R11	P005	P105	N7	NOR_INT_N
U12	P006	P106	N6	SD_PWR_REQ
T12	P007	P107	N5	AUDIO_AUX_REQ
U11	P008	P108	D8	
P11	P009	P109	D7	
P10	P010	P110	C8	
N10	P011	P111	E8	
P9	P012	P112	A4	SDRAM_DQ3
N9	P013	P113	A2	SDRAM_DQ4
T8	P014	P114	B3	SDRAM_DQ5
U8	P015	P115	A3	SDRAM_DQ6

U1C R7KA8P1KFLCAC#UC0

GPIO P2 / P3

C5	P200	P300	B5	SDRAM_DQ2
CEU_D0	B15	P201	P301	C4	SDRAM_DQ1
A17	P206	P302	A5	SDRAM_DQ0
	P207	P303_IN	B6	POWER_BUTTON_N
		P304	A10	
		P305	C10	
		IRQ28-DS	D11	SENS_INT1
		P307_IN	C11	VOL_UP_N
		P308	C9	
		P309_IN	E12	PAGE_PREV_N
		P310_IN	E10	PAGE_NEXT_N
		P311_IN	B12	VOL_DOWN_N
		P312	C13	

U1D R7KA8P1KFLCAC#UC0

GPIO P4 / P5

SD_CLK	P17	P400	P500	U7	
SD_CMD	N17	P401	P501	R8	
SD_DAT0	L14	P402	P502	T7	
SD_DAT1	H13	P403	P503	H2	SDRAM_A4
SD_DAT2	J13	P404	P504	H1	SDRAM_A5
SD_DAT3	G12	P405	P505	H3	SDRAM_A6
SD_CD_N	F14	P406	P506	J1	SDRAM_A7
SD_READY	R11	P407	P507	J2	SDRAM_A8
	M15	P408	P508	J3	SDRAM_A9
SENS_SDA	R16	SDA0_A	P509	J4	SDRAM_A10
SENS_SCL	K13	SCL0_A	P510	K3	SDRAM_A11
	M14	P411	P511	D15	CAM_SDA
COVER_INT_N	M14	P412	P512	D13	CAM_SCL
	P14	P413	P513	R13	
	L13	P414	P514	R14	
	R15	P415	P515	P15	

SDRAM_A[0..12]
SDRAM_BA[0..1]
NOR_DQ_HOST[0..7]
SDRAM_DQM[0..3]
SDRAM_DQ[0..31]

U1E R7KA8P1KFLCAC#UC0

GPIO P6 / P7

	P3	P600	P700	F12	
RADIO_SCLK	P4	P601	P701	F15	
RADIO_CIPO	P2	P602	P702	E13	
RADIO_COPI	P1	P603	P703	G14	CEU_D7
RADIO_CS_N	N2	P604	P704	G13	RADIO_DATA_READY
	N1	P605	P705	F17	RADIO_HANDSHAKE
	N3	P606	P706	E17	RADIO_RESET_REQ_N
SDRAM_DQ31	M2	P607	P707	F16	RADIO_PWR_EN
SDRAM_A12	K2	P608	P708	N12	SD_IO_REQ
SDRAM_DQ7	A1	P609	P709	P16	CAM_PWR_REQ
SDRAM_DQ12	D3	P610	P710	M13	
SDRAM_DQ13	D2	P611	P711	N15	
SDRAM_DQ14	E4	P612	P712	N13	
SDRAM_DQ15	C2	P613	P713	N16	
SDRAM_DQM0	E3	P614	P714	M16	
SDRAM_DQM2	E2	P615	P715	T15	

CMS-016: SCIO SCK=P601; CIPO=P602; COPI=P603; CS_N=P604.
READY=P704/IRQ26; HANDSHAKE=P705/IRQ19.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Timing, switched-domain isolation and reset arbitration remain open.

U1F R7KA8P1KFLCAC#UC0

GPIO P8 / P9

NOR_DQ_HOST5	T6	P800	P902	E9	CEU_D1
NOR_DS_HOST	P6	P801	P903_OD	D9	POWER_KILL_N
NOR_DQ_HOST6	R6	P802	P904	A16	AUDIO_AUX_PG
NOR_DQ_HOST1	P7	P803	P905	A14	CEU_D6
NOR_DQ_HOST7	R7	P804	P906	A13	CEU_D5
	P12	P805	P907	A15	CEU_D4
NOR_CK_HOST	U5	P806	P908	B13	CEU_D3
	T5	P807	P909_IN	B14	CEU_D2
	M9	P808	P910	E11	
	N8	P809	P911	C12	
	P8	P810	P912	D12	
SDRAM_CS_N	B1	P811	P913	E12	
		P812	P914	E5	
		P813	P915	A6	

U1G R7KA8P1KFLCAC#UC0

GPIO PA / PB

SDRAM_A3	G1	PA00	PB00	E16	
SDRAM_A2	H4	PA01	PB01	D17	
SDRAM_A1	F1	PA02	PB02	E13	CEU_VSYNC
SDRAM_A0	G2	PA03	PB03	D16	CEU_HREF
SDRAM_DQM3	D1	PA04	PB04	E13	CEU_PCLK
SDRAM_DQM1	G3	PA05	PB05	D15	
SDRAM_CKE	C1	PA06	PB06	E14	
	G4	PA07	PB07	D14	
SDRAM_WE_N	F3	PA08			
SDRAM_CAS_N	F4	PA09			
SDRAM_RAS_N	F2	PA10			
SDRAM_DQ8	B4	PA11			
SDRAM_DQ9	B2	PA12			
SDRAM_DQ10	C3	PA13			
SDRAM_DQ11	D4	PA14			
	E1	PA15			

SDRAM_CLK R43 0R SDRAM_CLK_SRC

U1H R7KA8P1KFLCAC#UC0

GPIO PC / PD

SDRAM_DQ30	M1	PC00	PD00	K4	SDRAM_BA0
SDRAM_DQ29	M3	PC01	PD01	B16	
SDRAM_DQ28	L2	PC02	PD02	B17	
SDRAM_DQ27	L1	PC03	PD03	C15	
SDRAM_DQ26	L3	PC04	PD04	C14	
SDRAM_DQ25	L5	PC05	PD05	C16	
SDRAM_DQ24	N4	PC06	PD06	C17	
SDRAM_DQ23	K5	PC07	PD07	E15	
SDRAM_DQ22	M4	PC08			
SDRAM_DQ21	L4	PC09			
SDRAM_DQ20	M5	PC10			
SDRAM_DQ19	H5	PC11			
SDRAM_DQ18	G5	PC12			
SDRAM_DQ17	J5	PC13			
SDRAM_DQ16	F5	PC14			
SDRAM_BA1	K1	PC15			

CMS-010 | R43: 0R source clock link.
SI tuning position; damping and timing remain unqualified.

U1M R7KA8P1KFLCAC#UC0

USB / MIPI

+3V3_USBHS_A	K15	AVCC_USBHS	AVCC_MIPI	T4	
	U16	PB14/USB_DP	MIPI_CL_N	U2	CAM_CLK_N
	T16	PB15/USB_DM	MIPI_CL_P	T2	CAM_CLK_P
	H17	USBHS_DM	MIPI_DL0_N	U1	CAM_D0_N
	H16	USBHS_DP	MIPI_DL0_P	T1	CAM_D0_P
USBHS_RREF	J15	USBHS_RREF	MIPI_DL1_N	U3	CAM_D1_N
+3V3_MCU	U7	VCC_USB	MIPI_DL1_P	T3	CAM_D1_P
+3V3_MCU	H15	VCC_USBHS	VCC18_MIPI	R2	
	T17	VSS_USB	VSS_MIPI	R3	
	J17	VSS1_USBHS			
	J16	VSS2_USBHS			

+3V3_MCU
C39 100n
+1V8_MIPI
GND
C111 100n
USBHS_RREF
R6 2.2k 1%

CMS-013: R2 VCC18_MIPI = +1V8_MIPI from U23.
C111: 100 nF local bypass to VSS_MIPI/GND
(RA8P1 QDG Rev.1.10, Table 1): C39 bypasses AVCC_MIPI.
Camera lanes and operating sequence remain pending.
Keep CSI/DSI stopped until integration is qualified.

USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md

+3V3_MCU C40 100n GND

USB-002: USB full-speed supply / U1.U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md

+3V3_MCU C41 10n GND

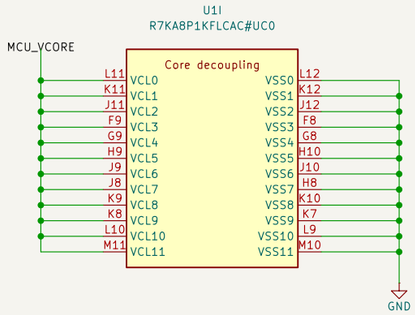
+3V3_MCU C42 47u 6.3V GND

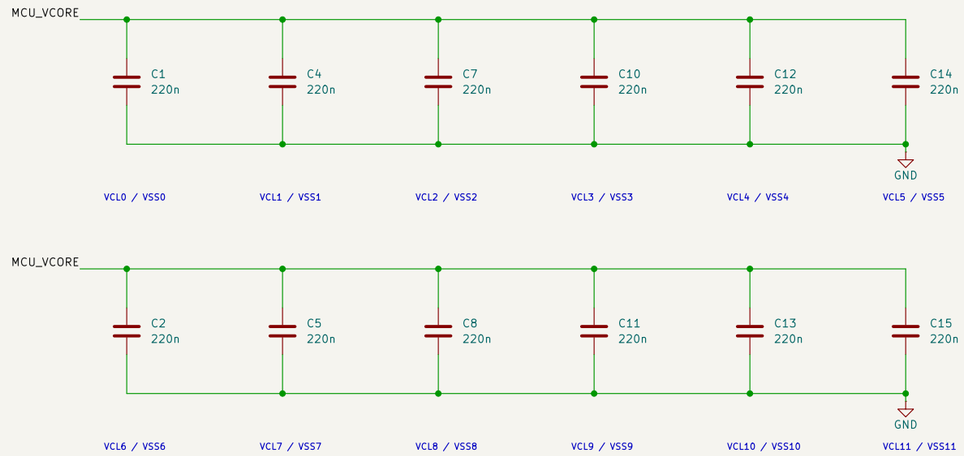
USB-003: USB-HS supply / U1.H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1+/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1+/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND: no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader
Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch
Title: RA8P1 IO allocation
Size: A2 | Date: 2026-09-05 | Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6 | Id: 2/15

CORE DECOUPLING — 220 nF AT EACH VCL / VSS PAIR

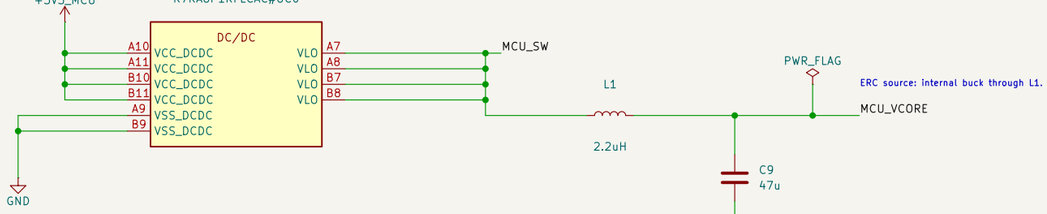


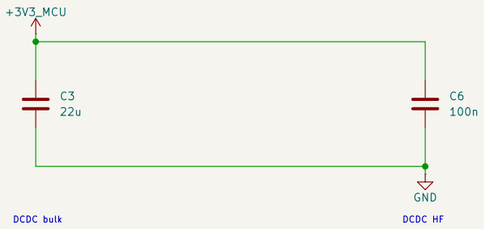


OFS2.DCDCEN = 1. All VLO balls share MCU_SW.
L1 feeds every VCL ball; never tie MCU_VCORE to 3.3 V.
C9 returns to VSS_DCDC. VCC_DCDC is tied to VCC.
HUM Rev.1.30, Table 69.2 / Fig.69.1, pp.4042–4044.

PWR-005 | LOCAL VCL BYPASS — C1,C2,C4,C5,C7,C8,C10–C15
220nF per VCL/VSS pair: Renesas QDG Rev1.10 section1.2 (p.8).
TDK C1608X7R1H224K080AB: 50V X7R +/−10%, twelve local capacitors.
C_initial = 220*(1 +/− .10) = 198..242nF each.
Bank = 12*220nF = 2.64uF nominal; 2.376..2.904uF initial.
TDK nominal curve: C(1.0V) = 220+(1/1.25)*(223.065−220) = 222.452nF.
Curve is reference only; bias/temperature/aging and PCB PDN remain unqualified.
Full Python-verified calculations and placement map: PWR-005, power_decoupling.md.

INTERNAL BUCK REGULATOR

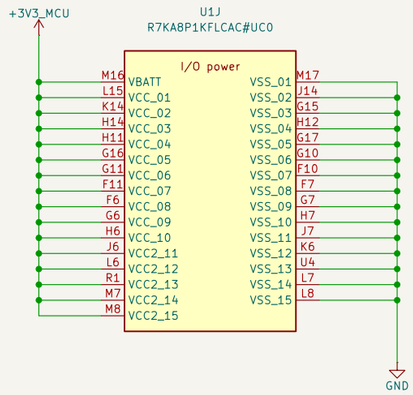


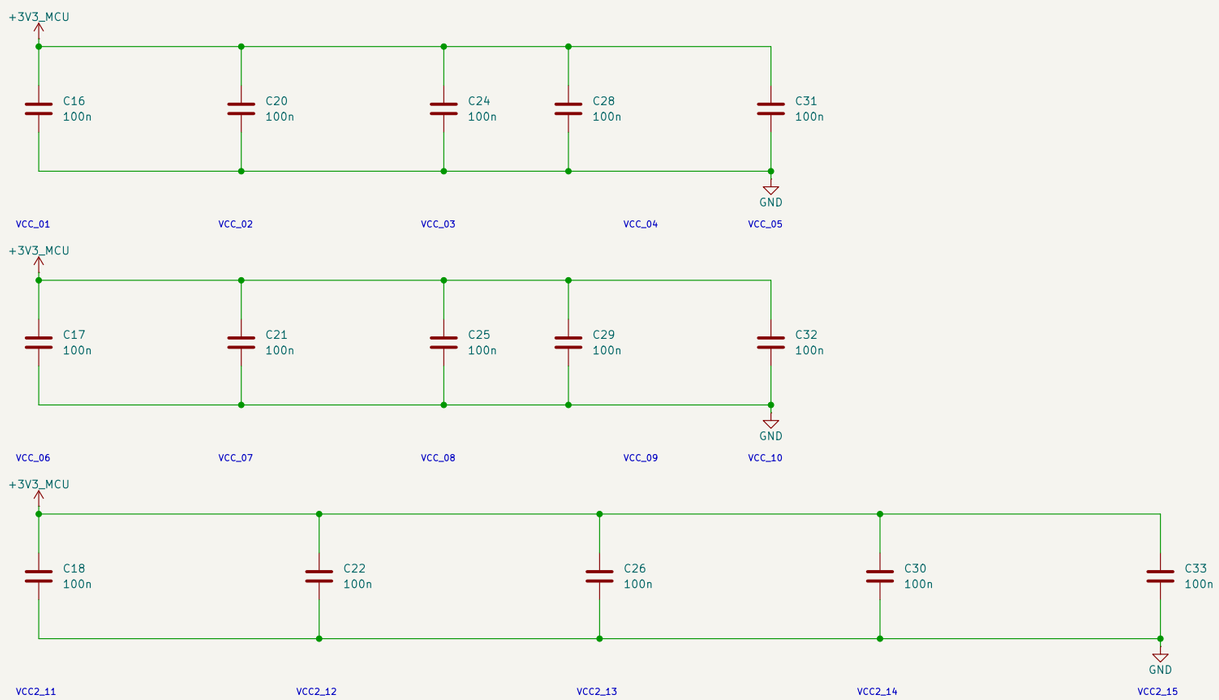


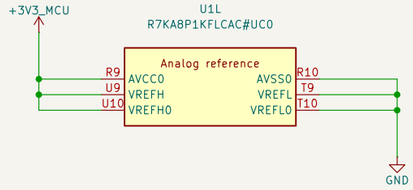
L1 DCR <=100mOhm. Effective capacitance and inductor current rating require final MPN qualification. +3V3_MCU is supplied by U13 on Main 3V3 digital supply; upstream SYS_AON source remains open.

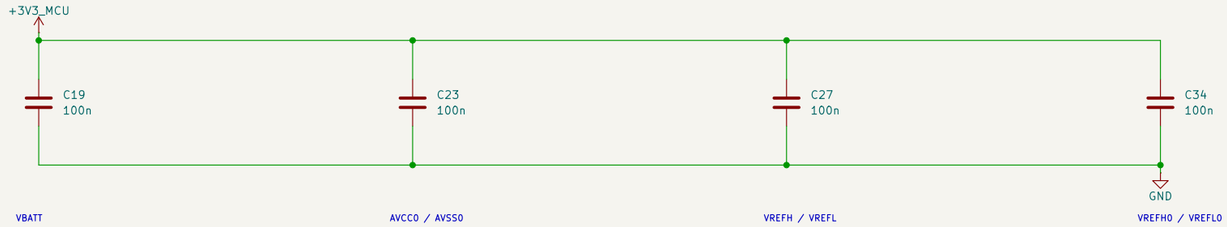
RA8P1 e-reader
Sheet: /RA8P1 internal core regulator/
File: mcu_core_power.kicad_sch
Title: RA8P1 internal core regulator
Size: A3 | Date: 2026-09-05 | **Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6 | Id: 3/15

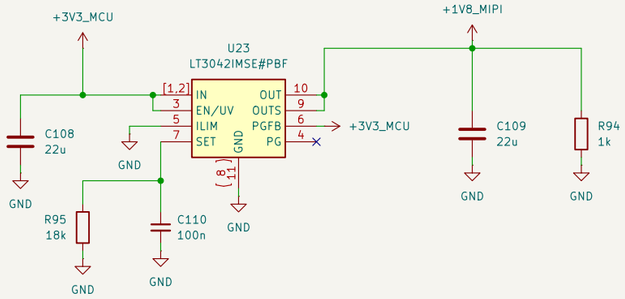
I/O DECOUPLING — 100 nF AT EACH NUMBERED VCC / VSS PAIR











VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB and MIPI supply pins and local bypasses are on the interface sheet.
U23 supplies VCC18_MIPI; camera lanes and sequencing remain pending.
See CMS-013 and design/power_decoupling.md for source-backed conditions.
This is an electrical draft; installed capacitor and PDN qualification remain open.

CMS-013 sourcing / value review — 2026-10-05
U23 LT3042 remains installed; circuit qualification open.
LT3060-1.8 candidate: $4.79 qty-one CT, reverse protected.
Existing 100n implies ~60ms typical startup: do not reuse blindly.
Startup/reset, PHY noise and native integration remain HOLD.
See design/mipi_ldo_selection.md; no purchase release.

CMS-013: LT3042 MIPI host 1.8 V supply — conditional design screens
R95: 18k, 0.1%, 25ppm/C; ISET 98..102uA, offset +/-2mV, leakage allocation 100nA.
Modeled DC: 1.754036706..1.846236894 V; RA8P1 range 1.65..1.95 V.
R94 1k permanent load: minimum 1.617488482mA; maintains >=1mA DC test load.
C110 100nF C0G: conditional minimum 94.715nF; SET slope 0.0010777971V/us.
SET charging model is not a guaranteed OUT ramp; RA8P1 minimum rise 8.4us/V.
C108/C109: 22uF * 0.8 * 0.85 * 0.6 = 8.976uF each; >=4.7uF required.
60% retained capacitance and 100nA leakage allocations require verification.
EN and PGFB follow IN; fast start disabled. ILIM grounded; PG unused.
ADI LT3042 Rev C; RA8P1 datasheet Tables 2.2, 2.39, 2.44. Python checked.
See design/camera_storage_interfaces.md CMS-013 for sources and full conditions.
Startup/restart, rail sequencing/discharge, installed capacitance and PDN remain open.

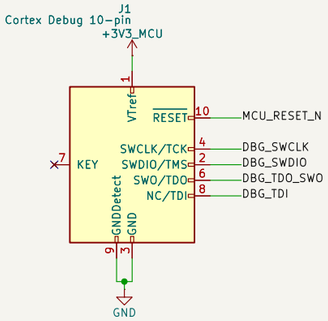
CMS-013: +20mA steady CSI branch planning allocation
4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
+ 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
13mA is an engineering envelope, not an interpolated guarantee.
Auxiliary allowance requires verification; startup/inrush excluded.
AVCC_MIPI remains within existing MCU allocation.
Main subtotal: 2.075 + 0.020 = 2.095A; full system budget pending.
At 3.393012496V, 3.2V input and assumed 75% efficiency:
input 2.961817158A; converter loss 2.369453727W.
Source, current and thermal acceptance remain open.
Sources, conditions and Python: design/camera_storage_interfaces.md.

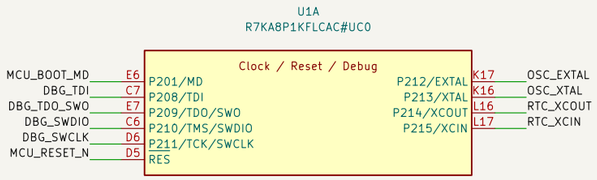
RA8P1 e-reader

Sheet: /RA8P1 IO and analog supplies/
File: mcu_supply_power.kicad_sch

Title: RA8P1 I/O and analog supplies

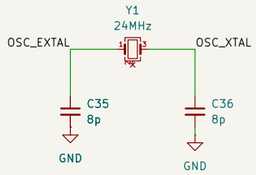
Size: A3	Date: 2026-09-05	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/15

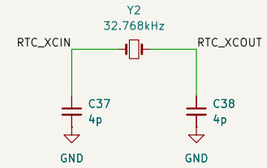


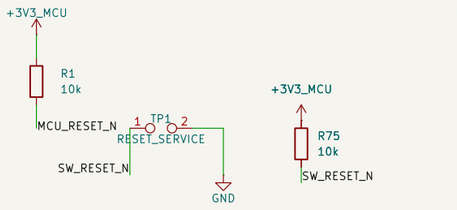


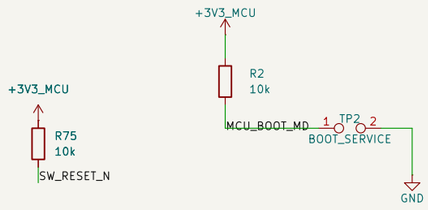
CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF: effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

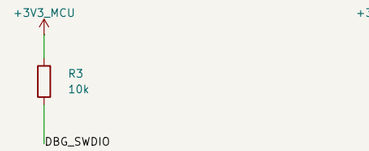
CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF: Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md

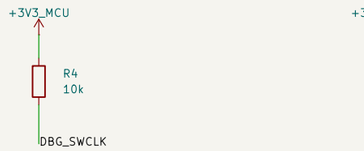


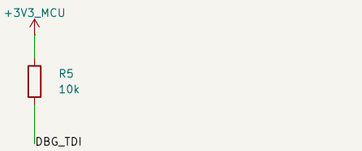


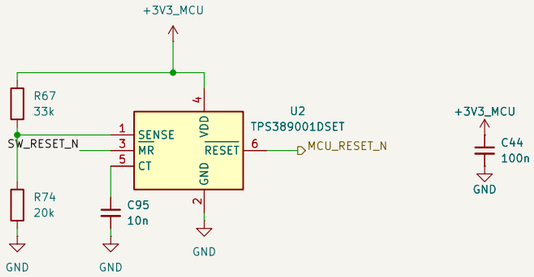












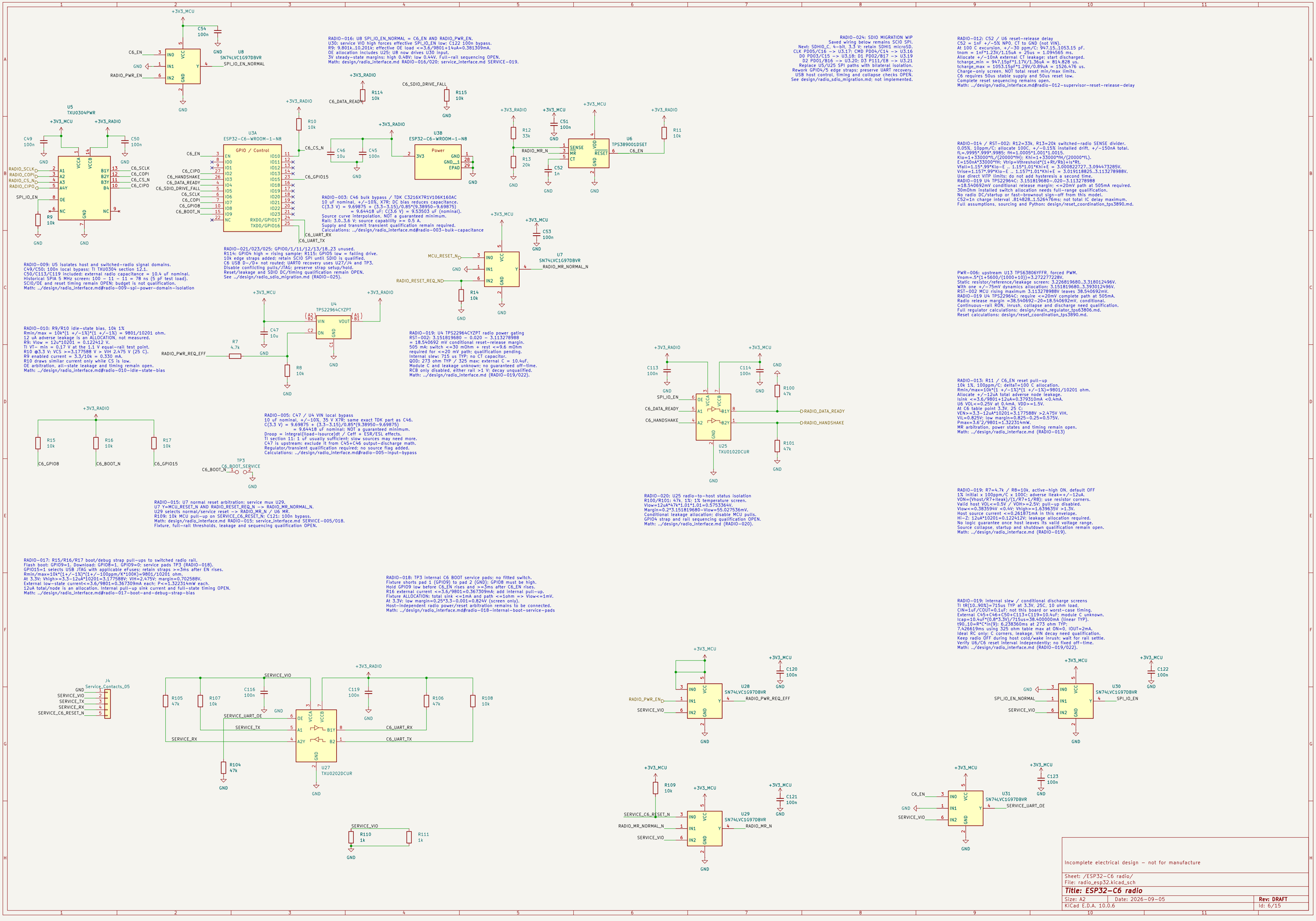
RST-002: U2 TPS389001DSET; R67=33k, R74=20k, R75=10k, C95=10n.
Sense resistors: +/-0.05%, 10ppm/C over 100C, allocated +/-0.15% assembly/aging.
fL=.9995*.999*.9985=.99700274925; fH=1.0005*1.001*1.0015=1.00300275075.
Klo=1+33000*fL/(20000*fH); Khi=1+33000*fH/(20000*fL); E=150nA*33000*fH.
Vfall=1.15*.99*Klo-E .. 1.15*1.01*Khi+E = 3.000822727..3.094473285V.
Vrise=1.157*.99*Klo-E .. 1.157*1.01*Khi+E = 3.019118825..3.113278988V.
Main release margin=3.151819680-3.113278988=38.540692mV; no double hysteresis.
C95: 10n C0G +/-5%, 30ppm/C over 100C; allocate +/-10nA external CT leakage.
tcharge,min=10n*.95*.997*1.17/(1.35u+10n)=8.148276ms > 2.4ms RES minimum.
tcharge,max=10n*1.05*1.003*1.29/(.90u-10n)=15.264758ms; NOT total-delay max.
R75 MR pull-up: R=9801..10201ohm; allocate +/-50uA total adverse MR current.
MR high margin=.3*3.151819680-50uA*10201=.435495904V; sink <=.396190439mA.
Conditional DC/startup design, not fast-collapse or SDRAM-quiescence proof.
Full derivation, assumptions, sourcing and Python: design/reset_coordination_tps3890.md.

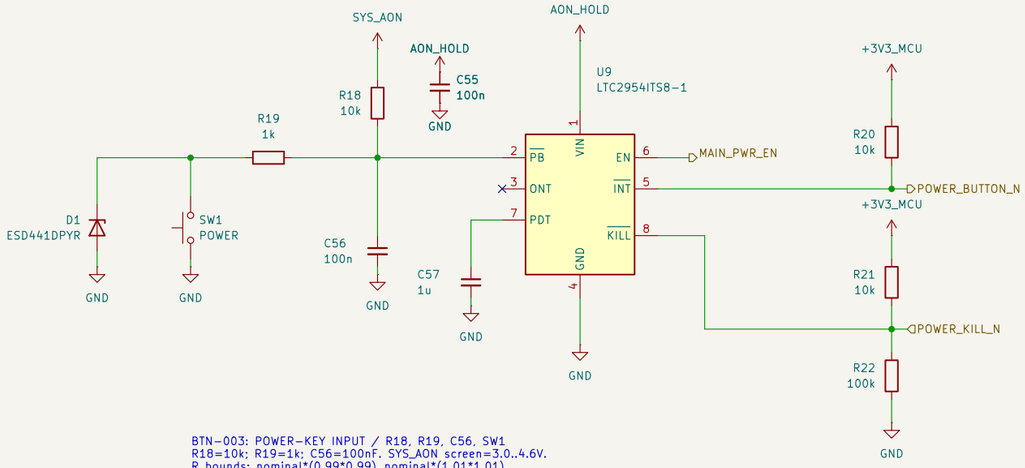
References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

Sheet: /RA8P1 clocks, reset and debug/
File: mcu_clocks_debug.kicad_sch

Title: RA8P1 clocks, reset and debug

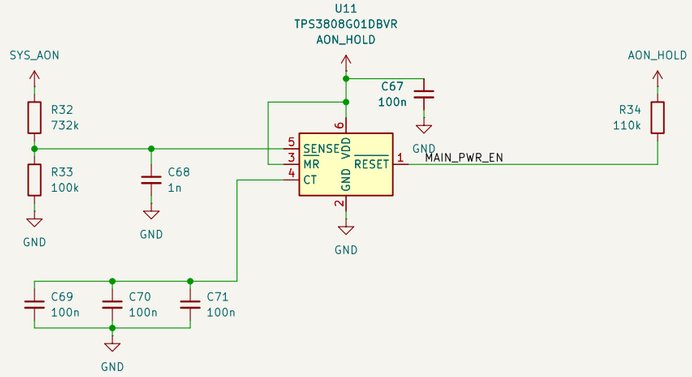
Size: A3	Date: 2026-09-05	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 5/15


+3V3_MCU
C54
100n
GND
U8
SN74LVC1G97DBVR
C6_EN
GND
RADIO_PWR_EN
SPI_IO_EN_NORMAL
IN0
IN1
IN2
VCC
GND
Y
RADIO-016: U8 SPI_IO_EN_NORMAL = C6_EN AND RADIO_PWR_EN.
U30: service VIO high forces effective SPI_IO_EN low; C122 100n bypass.
R9: 9.801k..10.201k; effective OE load <=3.6/9801+14uA=0.381309mA.
OE allocation includes U25; U8 now drives U30 input.
3V steady-state margins: high 0.48V; low 0.44V. Full-rail sequencing OPEN.
Math: design/radio_interface.md RADIO-016/020; service_interface.md SERVICE-019.
RADIO-024: SDIO MIGRATION WIP
Saved wiring below remains SCIO SPI.
Next: SDHIO_C, 4-bit, 3.3 V; retain SDHI1 microSD.
CLK PD05/C16 -> U3.17; CMD PD04/C14 -> U3.16
D0 PD03/C15 -> U3.18; D1 PD02/B17 -> U3.19
D2 PD01/B16 -> U3.20; D3 PC11/E8 -> U3.21
Replace U5/U25 SPI paths with bilateral isolation.
Rework GPIO4/5 edge straps; preserve UART recovery.
USB host control, timing and collapse checks OPEN.
See design/radio_sdio_migration.md; not implemented.
RADIO-012: C52 / U6 reset-release delay
C52 = 1nF +/-5% NPO, CT to GND (not VIN).
At 100 C excursion, +/-30 ppm/C: 947.15..1053.15 pF.
tnom = 1nF*1.23V/1.15uA = 25us = 1.094565 ms.
Allocate +/-10nA external CT leakage; start discharged.
tcharge_min = 947.15pF*1.17V/1.36uA = 814.828 us.
tcharge_max = 1053.15pF*1.29V/0.89uA = 1526.476 us.
Charge-only screen, NOT total reset min/max limits.
C6 requires 50us stable supply and 50us reset low.
Complete reset sequencing remains open.
Math: ../design/radio_interface.md#radio-012-supervisor-reset-release-delay
U5
TXU0304PWR
+3V3_MCU
+3V3_RADIO
C49
100n
C50
100n
GND
RADIO_SCLK
RADIO_COPI
RADIO_CS_N
RADIO_CIPO
SPI_IO_EN
R9
10k
C6_SCLK
C6_COPI
C6_CS_N
C6_CIPO
A1
A2
A3
A4Y
B1Y
B2Y
B3Y
B4
OE
NC
VCCA
VCCB
U3A
ESP32-C6-WROOM-1-N8
GPIO / Control
C6_EN
C6_CIPO
C6_HANDSHAKE
C6_DATA_READY
C6_SDIO_DRIVE_FALL
C6_SCLK
C6_COPI
C6_GPIO8
C6_BOOT_N
C6_GPIO15
C6_UART_RX
C6_UART_TX
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
NC
IO10
IO11
IO12
IO13
IO14
IO15
IO18
IO19
IO20
IO21
IO22
IO23
RXD0/GPIO17
TXD0/GPIO16
R10
10k
C6_CS_N
C6_GPIO15
RADIO-003: C46 bulk bypass / TDK C3216X7R1V106K160AC
10 uF nominal, +/-10%, X7R: DC bias reduces capacitance.
C(3.3 V) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)
= 9.64418 uF; C(3.6 V) = 9.53503 uF (nominal).
Source curve interpolation, NOT a guaranteed minimum.
Rails: 3.0..3.6 V; source capability >= 0.5 A.
Supply and transmit transient qualification remain required.
Calculations: ../design/radio_interface.md#radio-003-bulk-capacitance
U3B
ESP32-C6-WROOM-1-N8
Power
3V3
GND
GND__1
EPAD
C46
10u
C45
100n
R114
10k
C6_DATA_READY
R115
10k
C6_SDIO_DRIVE_FALL
R12
33k
R13
20k
RADIO_MR_N
C51
100n
C52
1n
U6
TPS389001DSET
SENSE
MR
CT
VDD
RESET
C6_EN
R11
10k
RADIO-014 / RST-002: R12=33k, R13=20k switched-radio SENSE divider.
0.05%, 10ppm/C; allocate 100C, +/-0.15% installed drift, +/-150nA total.
fl=.9995*.999*.9985: fH=1.0005*1.001*1.0015.
Klo=1+33000*fL/(20000*fH); Khi=1+33000*fH/(20000*fL).
E=150nA*33000*fH; Vtrip=Vthreshold*(1+R1/R2)+Is*R1.
Vfall=1.15*.99*Klo-E .. 1.15*1.01*Khi+E = 3.000822727..3.094473285V.
Vrise=1.157*.99*Klo-E .. 1.157*1.01*Khi+E = 3.019118825..3.113278988V.
Use direct VITP limits; do not add hysteresis a second time.
RADIO-019 U4 TPS22964C: 3.151819680-.020-3.113278988
=18.540692mV conditional release margin: <=20mV path at 505mA required.
30mOhm installed switch allocation needs full-range qualification.
No radio DC/startup or fast-brownout sign-off from this model.
C52=1n charge interval .814828..1.526476ms; not total IC delay maximum.
Full assumptions, sourcing and Python: design/reset_coordination_tps3890.md.
RADIO-009: U5 isolates host and switched-radio signal domains.
C49/C50: 100n local bypass: TI TXU0304 section 12.1.
C50/C113/C119 included; external radio capacitance = 10.4 uF nominal.
Historical SPIA 5 MHz screen: 100 -- 11 -- 11 = 78 ns (5 pF test load).
SCIO/OE and reset timing remain OPEN: budget is not qualification.
Math: ../design/radio_interface.md#radio-009-spi-power-domain-isolation
RADIO-021/023/025: GPIO0/1/11/12/13/18..23 unused.
R114: GPIO4 high = rising sample; R115: GPIO5 low = falling drive.
10k edge straps added; retain SCIO SPI until SDIO is qualified.
C6 USB D-/D+ not routed; UART0 recovery uses U27/J4 and TP3.
Disable conflicting pulls/JTAG; preserve strap setup/hold.
Reset/leakage and SDIO DC/timing qualification remain OPEN.
See ../design/radio_sdio_migration.md (RADIO-025).
U7
SN74LVC1G97DBVR
MCU_RESET_N
RADIO_RESET_REQ_N
RADIO_MR_NORMAL_N
R14
10k
C53
100n
RADIO-015: U7 normal reset arbitration; service mux U29.
U7 Y=MCU_RESET_N AND RADIO_RESET_REQ_N -> RADIO_MR_NORMAL_N.
U29 selects normal/service reset -> RADIO_MR_N / U6 MR.
R109: 10k MCU pull-up on SERVICE_C6_RESET_N; C121: 100n bypass.
Math: design/radio_interface.md RADIO-015; service_interface.md SERVICE-005/018.
Fixture, full-rail thresholds, leakage and sequencing qualification OPEN.
PWR-006: upstream U13 TPS63806YFFR, forced PWM.
Vnom=.5*(1+5600/(1000+10))=3.272277228V.
Static resistor/reference/leakage screen: 3.226819680..3.318012496V.
With one +/-75mV dynamics allocation: 3.151819680..3.393012496V.
RST-002 MCU rising maximum 3.113278988V leaves 38.540692mV.
RADIO-019 U4 TPS22964C: require <=20mV complete path at 505mA.
Radio release margin =38.540692-20=18.540692mV, conditional.
Continuous-rail RON, inrush, collapse and discharge need qualification.
Full regulator calculations: design/main_regulator_tps63806.md.
Reset calculations: design/reset_coordination_tps3890.md.
RADIO-010: R9/R10 idle-state bias, 10k 1%
Rmin/max = 10k*(1 +/-1%)*(1 +/-1%) = 9801/10201 ohm.
12 uA adverse leakage is an ALLOCATION, not measured.
R9: Vlow = 12u*10201 = 0.122412 V.
TI VT- min = 0.17 V at the 1.1 V equal-rail test point.
R10 @3.3 V: VCS >=3.177588 V > VIH 2.475 V (25 C).
R9 enabled current = 3.3/10k = 0.330 mA.
R10 draws similar current only while C5 is low.
OE arbitration, all-state leakage and timing remain open.
Math: ../design/radio_interface.md#radio-010-idle-state-bias
U4
TPS22964CYZPT
VIN
VOUT
ON
GND
C47
10u
C2
C1
R7
4.7k
R8
10k
RADIO_PWR_REQ_EFF
RADIO-019: U4 TPS22964CYZPT radio power gating
RST-002: 3.151819680 - 0.020 - 3.113278988
= 18.540692 mV conditional reset-release margin.
505 mA; switch <=30 mOhm + rest <=9.6 mOhm
required for <=20 mV path; qualification pending.
Internal slew: 715 us TYP; no CT capacitor.
QOD: 273 ohm TYP / 325 max; external C = 10.4uF.
Module C and leakage unknown; no guaranteed off-time.
RCB only disabled, either rail >1 V; decay unqualified.
Math: ../design/radio_interface.md (RADIO-019/022).
RADIO-013: R11 / C6_EN reset pull-up
10k 1%, 100ppm/C; deltaT=100 C allocation.
Rmin/max=10k*(1 +/-1%)*(1 +/-1%)=9801/10201 ohm.
Allocate +/-12uA total adverse node leakage.
Isink <=3.6/9801+12uA=0.379310mA <0.4mA.
U6 VOL<=0.25V at 0.4mA, VDD>=1.5V.
At C6 table point 3.3V, 25 C:
VEN>=3.3-12uA*10201=3.177588V >2.475V VIH.
VIL=0.825V; low margin=0.825-0.25=0.575V.
Pmax=3.6^2/9801=1.322314mW.
MR arbitration, power states and timing remain open.
Math: ../design/radio_interface.md (RADIO-013)
R15
10k
R16
10k
R17
10k
C6_GPIO8
C6_BOOT_N
C6_GPIO15
RADIO-005: C47 / U4 VIN local bypass
10 uF nominal, +/-10%, 35 V X7R: same exact TDK part as C46.
C(3.3 V) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)
= 9.64418 uF nominal; NOT a guaranteed minimum.
Droop = integral(iload-isource)dt / Ceff + ESR/ESL effects.
TI section 11: 1 uF usually sufficient; slow sources may need more.
C47 is upstream: exclude it from C45+C46 output-discharge math.
Regulator/transient qualification required; no source flag added.
Calculations: ../design/radio_interface.md#radio-005-input-bypass
TP3
C6_BOOT_SERVICE
C6_BOOT_N
U25
TXU0102DCUR
C113
100n
C114
100n
R100
47k
R101
47k
SPI_IO_EN
C6_DATA_READY
C6_HANDSHAKE
RADIO_DATA_READY
RADIO_HANDSHAKE
RADIO-020: U25 radio-to-host status isolation
R100/R101: 47k, 1%; 1% temperature screen.
Vlow=12uA*47k*1.01*1.01=0.5753364V.
Margin=0.2*3.15181968O-Vlow=55.027636mV.
Conditional leakage allocation; disable MCU pulls.
GPIO4 strap and rail sequencing qualification OPEN.
Math: ../design/radio_interface.md (RADIO-020).
RADIO-019: R7=4.7k / R8=10k, active-high ON, default OFF
1% initial x 100ppm/C x 100C; adverse ileak=+/-12uA.
VON=(Vhost/R7+Ileak)/(1/R7+1/R8); use resistor corners.
Valid host VOL<=0.5V / VOH>=2.5V: pull-up disabled.
Vlow<=0.38359V -0.4V; Vhigh>=1.639635V >1.3V.
Host source current <=0.261871mA in this envelope.
Hi-Z: 12uA*10201=0.122412V: leakage allocation required.
No logic guarantee once host leaves its valid voltage range.
Source collapse, startup and shutdown qualification remain open.
Math: ../design/radio_interface.md (RADIO-019).
RADIO-017: R15/R16/R17 boot/debug strap pull-ups to switched radio rail.
Flash boot: GPIO9=1, Download: GPIO8=1, GPIO9=0; service pads TP3 (RADIO-018).
GPIO15=1 selects USB JTAG with applicable eFuses; retain straps >=3ms after EN rises.
Rmin/max=10k*(1+/-1%)*(1+/-100ppm/K*100K)=9801/10201 ohm.
At 3.3V: Vhigh>=3.3-12uA*10201=3.177588V; VIH=2.475V; margin=0.702588V.
External low-state current<=3.6/9801=0.367309mA each; P<=1.322314mW each.
12uA total/node is an allocation. Internal pull-up sink current and full-state timing OPEN.
Math: ../design/radio_interface.md#radio-017-boot-and-debug-strap-bias
RADIO-018: TP3 internal C6 BOOT service pads; no fitted switch.
Fixture shorts pad 1 (GPIO9) to pad 2 (GND): GPIO8 must be high.
Hold GPIO9 low before C6_EN rises and >=3ms after C6_EN rises.
R16 external current <=3.6/9801=0.367309mA: add internal pull-up.
Fixture ALLOCATION: total sink <=1mA and path <=1ohm => Vlow<=1mV.
At 3.3V: low margin=0.25*3.3-0.001=0.824V (screen only).
Host-independent radio power/reset arbitration remains to be connected.
Math: ../design/radio_interface.md#radio-018-internal-boot-service-pads
RADIO-019: internal slew / conditional discharge screens
TI tR(10..90%)=715us TYP at 3.3V, 25C, 10 ohm load.
CIN=1uF/COUT=0.1uF: not this board or worst-case timing.
External C45+C46+C50+C113+C119=10.4uF: module C unknown.
Icap=10.4uF*(0.8*3.3V)/715us=38.400000mA (linear TYP).
t90..10=R*C*ln(9): 6.238360ms at 273 ohm TYP;
7.426619ms using 325 ohm table max at ON=0, IOUT=2mA.
Ideal RC only; C corners, leakage, VIN decay need qualification.
Keep radio OFF during host cold/wake inrush; wait for rail settle.
Verify U6/C6 reset interval independently; no fixed off-time.
Math: ../design/radio_interface.md (RADIO-019/022).
J4
Service_Contacts_05
GND
SERVICE_VIO
SERVICE_TX
SERVICE_RX
SERVICE_C6_RESET_N
R105
47k
R107
10k
C116
100n
C119
100n
R106
47k
R108
10k
SERVICE_UART_OE
SERVICE_TX
SERVICE_RX
C6_UART_RX
C6_UART_TX
R104
47k
U27
TXU0202DCUR
U28
SN74LVC1G97DBVR
RADIO_PWR_EN
RADIO_PWR_REQ_EFF
C120
100n
U30
SPI_IO_EN_NORMAL
SPI_IO_EN
C122
100n
R109
10k
U29
SERVICE_C6_RESET_N
RADIO_MR_NORMAL_N
RADIO_MR_N
C121
100n
U31
C6_EN
C123
100n
R110
1k
R111
1k
Incomplete electrical design — not for manufacture
Sheet: /ESP32-C6 radio/
File: radio_esp32.kicad_sch
Title: ESP32-C6 radio
Size: A2
Date: 2026-09-05
Rev: DRAFT
KiCad E.D.A. 10.0.6
Id: 6/15



BTN-003: POWER-KEY INPUT / R18, R19, C56, SW1
R18=10k; R19=1k; C56=100nF. SYS_AON screen=3.0..4.6V.
R bounds: nominal*(0.99*0.99)..nominal*(1.01*1.01).
At PB=0.6V: Isource<=(4.6-0.6)/9801 + 15uA + 12uA
<435.13uA; Isink=0.6/(1020.1+0.5)>587.88uA.
Thus the closed switch pulls PB below its 0.6V falling threshold.
Min contact current=3/(10201+1020.1+0.5)-12uA>255.34uA.
Open contact>=2.87748V; SW1 minimum rating=10uA at 2V.
C56 discharge peak<=4.6/980.1<4.694mA < SW1 50mA rating.
12uA is a leakage ALLOCATION; ESD/transient validation required.
Full derivation + Python: ../design/single_button_power.md BTN-003/005/010.

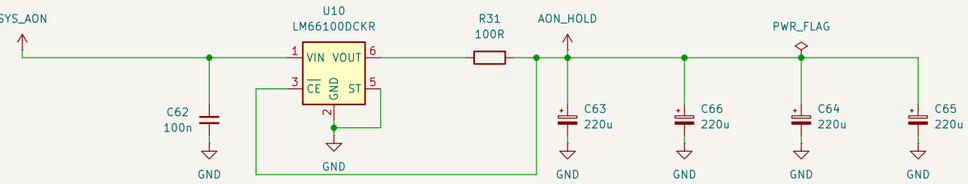
BTN-004: HARDWARE FORCED-OFF TIMING / C57
C57=1uF: TDK C2012X7R1E105K125AB, 25V X7R +/-10%.
C[uF]=1.56e-4*(t_added[ms]-1).
t_force,nom=64ms + 1ms + 1/1.56e-4=6475.256ms (6.48s).
C tolerance/temp only: 1*0.90*0.85=0.765uF;
1*1.10*1.15=1.265uF -> 4.969..8.174s using NOMINAL IC equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
ONT intentionally open: internal turn-on debounce=26..41ms.
Release the initial ON press; after forced off, release then press again.
Full calculation + Python: ../design/single_button_power.md BTN-004/005.



SYS-007 / SENS-010: SHUTDOWN / U12, Q1-Q3, R35-R40, R70-R71
POWER_OFF_H=NOT(MAIN_PWR_EN): held-domain VCC=2.7..4.6V.
Each gate branch: 1k series / 1M pulldown; 1% +100ppm/C, 100C.
4-gate peak<=4*4.6/(1000*0.99^2)=18.773595mA <50mA.
Includes future USB-clear; static<=22.773595uA <100uA.
VGSmin=(2.6-1020.1*1uA)/(1+1020.1/980100)=2.596278V.
1uA per gate/board leakage is a qualification allocation.
RUN/KILL/discharge/USB-clear FET drains remain separate.
R39||R40 + Q2: Rmax=22*1.01^2/2+0.2=11.4211ohm.
Ir<=2.5mA: 1.5mA key-filter return +1mA other injection.
Vinf=Ir*Rmax=28.552750mV; Cmain<=1mF.
Total=10ms+Rmax*Cmain*ln((3.6-Vinf)/(0.09-Vinf))+10ms
=66.398757ms <75.794309ms hold: margin9.395552ms.
Each 22R: P<=3.6^2/(22*0.99^2)=0.601052W: 1W at70C.
Qualify FET resistance, response, derating, backfeed and cycling.
Gate/transient charge shares 1uC reserve; allocations need validation.
VBATT firmware/option invariant remains unimplemented.
Sources/Python: ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

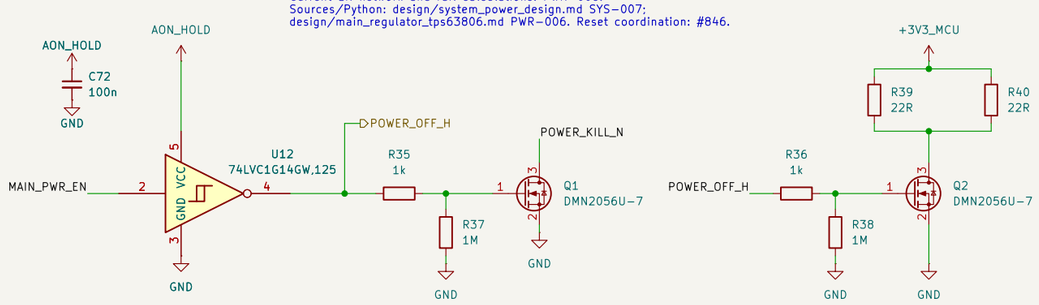
INCOMPLETE POWER CONTROL — NOT FOR FABRICATION
PB, INT, KILL, held supply and source supervisor are connected.
U12/Q1/Q2 provide fault KILL clamp and main-rail discharge.
MAIN_PWR_EN feeds raw U16; held Q3 separately clamps U13 EN.
SYS_AON source and USB permission-latch clear remain to be built.
Current converter: U13 TPS63806; design/main_regulator_tps63806.md PWR-006.
MCU reset: design/reset_coordination_tps3890.md RST-002.
Radio release margin and system shutdown coordination remain open (#846).

ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.

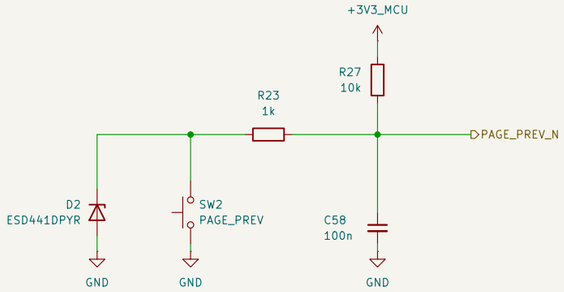


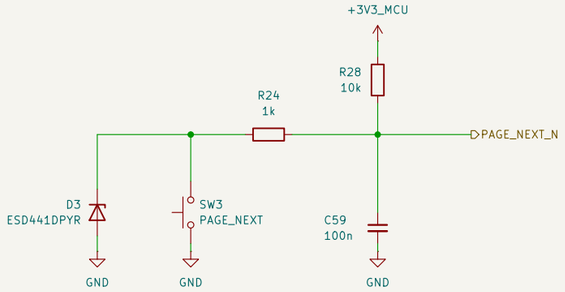
SYS-007 / SENS-010: HELD SUPPLY / U10, R31, C62-C66
C63-C66: 4 x 220uF/10V; Cmin=880uF*0.9^3=641.52uF.
VIN=raw SYS; VOUT->R31->reservoir; CE senses reservoir.
Icontrol<=1.825mA incl 1.100mA bank leakage screen.
Reverse allowance=0.817mA; Itotal<=2.642mA.
Vinitial=3.263705831-max(0.250,1.825mA*102.24ohm)
=3.013705831V; shared transition reserve=1uC.
thold=[641.52uF*(Vinitial-2.7V)-1uC]/2.642mA
=75.794309ms, conditional allocation.
Main C<=1mF, Rdis<=11.4211ohm, return/injection<=2.5mA:
10ms response +46.398757ms to90mV +10ms low hold
=66.398757ms total; hold margin=9.395552ms.
Recovery: raw>=3.70V, held>=3.25V; 1s settling PLUS
full source qualification; recharge model=0.657961s.
Qualify startup leakage, raw VIN=0, backfeed and cycling.
See ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

SYS-007 / PWR-006: SOURCE SUPERVISION / U11, R32-R34, C67-C71
Vtrip=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-2%, Isense +/-25nA; each R +/-0.1%, 25ppm/C, 100C.
Corner trip=3.263705831..3.476597632V.
C68: tau=(732k||100k)*1n=87.980769us (nominal).
C69-C71: 3x100n C0G; initial/temp=284.145..316.827nF.
CT delay=300/175+0.0005=1.714786s nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k: ENhigh>=2.7-3.8uA*110k*1.01^2=2.273598V.
POR sink=(1.3-0.2)/(110k*0.99^2)+3.8uA=14.003041uA.
MAIN_PWR_EN drives raw U16 buffer; held Q3 clamps TPS63806 EN.
Held U12 also drives KILL/discharge; USB-clear still needed.
Current EN network and rail calculations: PWR-006.
Sources/Python: design/system_power_design.md SYS-007;
design/main_regulator_tps63806.md PWR-006. Reset coordination: #846.



Sheet: /Power button and source control/		
File: power_button.kicad_sch		
Title: RA8P1 E-reader — power button and source control		
Size: A3	Date: 2026-09-07	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 7/15

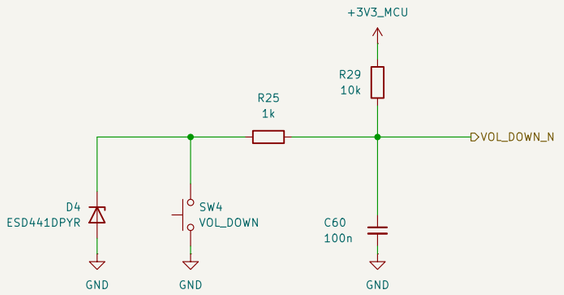


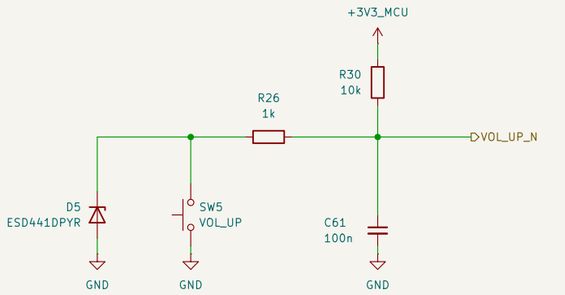


CMS-009 — FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5: R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU: 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=980.1..1020.1 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V: switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.





CMS-009 — FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

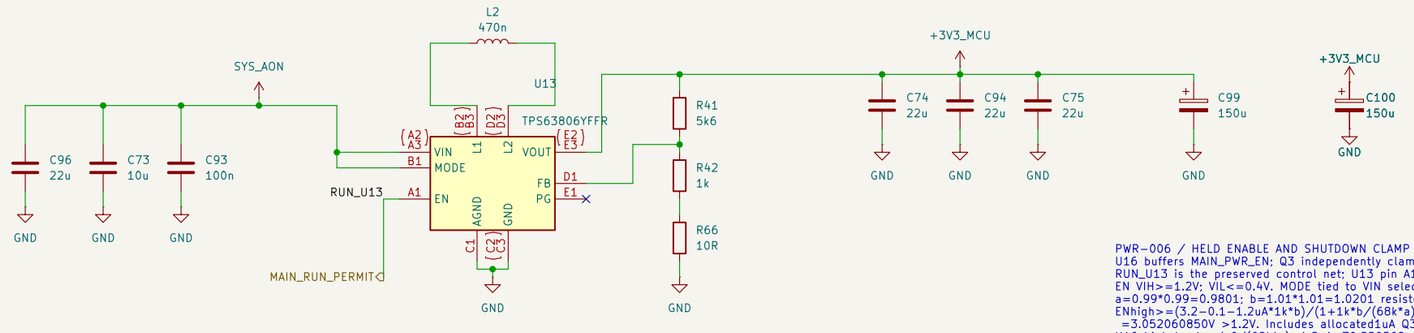
Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

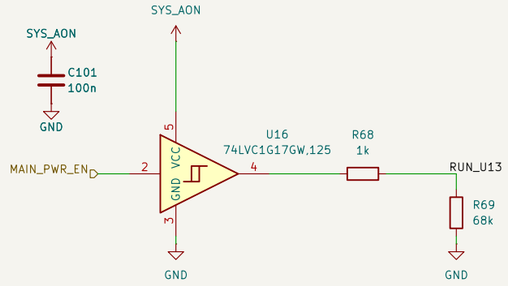
Sheet: /Page and volume buttons/
File: user_controls.kicad_sch

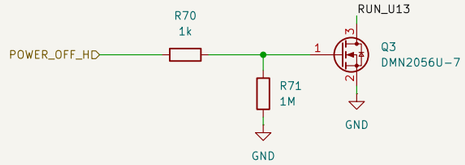
Title: RA8P1 E-reader — page and volume buttons

Size: A3	Date: 2026-09-07	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 8/15

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC - NOT FOR POWER-UP OR FABRICATION.







PWR-006 / FEEDBACK AND LOCAL PASSIVES
Vnom=0.500*(1+5600/(1000+10))=3.272277228 V.
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed bounds: initial*TCR(100C)*aging*assembly, then ohms.
R41=5600*(1+/-0.0002)*(1+/-0.0005)*(1+/-0.001)
*(1+/-0.0005)+/-0.02 = 5587.669237..5612.349243 ohm.
R42=1000*(1+/-0.0001)*(1+/-0.0002)*(1+/-0.001)
*(1+/-0.0005)+/-0.01 = 998.190970..1001.810970 ohm.
R66=10*(1+/-0.01)*(1+/-0.02)*(1+/-0.01)+/-0.05
=9.554980..10.455020 ohm; one combined aging/assembly 1%.
Vlo=0.495*(1+R41min/(R42max+R66max))-100nA*R41max.
Vhi=0.505*(1+R41max/(R42min+R66min))+100nA*R41max.
Vstatic=3.226819680..3.318012496 V (FPWM only).
One combined +/-75mV regulation/ripple/transient/routing allowance:
Vcomplete=3.151819680..3.393012496 V; qualification required.
SDRAM static VOH margin=2.4-0.7*3.393012496=24.891253mV.
This is not measured SI/noise headroom or reset qualification.

L2: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
=>0.3708..0.5698uH versus TI0.37..0.57uH. Qualify hot/current.
C74/C75/C94: 3*22uF*0.8*0.85*0.6=26.928uF > TI21uF.
C96: 22uF*0.8*0.85*0.6=8.976uF > TI4uF input minimum.
0.6 is allocated combined bias/aging retention, NOT guaranteed.
Samsung typical output-bias retention at3.393012496V=90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LOAD, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold-start screen, not measured load.
Pout=3.393012496*1.8=6.107422493W.
At allocated75% efficiency: Pin=8.143229991W; Iin@3.2V=2.544759372A.
Loss=2.035807498W: NOT an acceptable continuous thermal result.
At90% efficiency: loss=0.678602499W.
TJ screen=70C+78.8C/W*0.678602499W=123.474C.
TI board theta-JA is not the final PCB thermal resistance.
2.25A screen: Pin=3.393012496*2.25/0.75=10.179037489W.
Converter limit, source limit, startup and thermal envelope need validation.

PWR-006 storage checkpoint (nominal, not effective minimum):
Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
=479.61uF cold; radio-on adds10.2uF =>489.81uF.
C99/C100 remain150uF each. C94 adds22uF to the output.
Old bootstrap/private VCC/VC capacitors were not main-rail storage.
Whole-main ceiling remains1mF; reconcile later peripheral additions.
Cout lower acceptance and distributed PDN/startup are separate checks.

SENS-010 discharge revision: Rmax11.4211ohm; return/injection<=2.5mA.
Includes1.5mA key return +1mA other backfeed; Vinf=0.028552750V.
t=1mF*11.4211ohm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.398757ms total, below held75.794309ms screen.
Includes10ms command/clamp/discharge response and10ms below90mV.
Held budget:1.825mA control +0.817mA reverse;1uC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON source protection, loaded startup and fast faults remain open.
RST-002 / RADIO-019: release margin remains conditional on rail/path bounds.
Do not power up until reset coordination and source protection are complete.
Full assumptions: ../design/main_regulator_tps63806.md;
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.

PWR-006 / HELD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWR_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
a=0.99*0.99=0.9801; b=1.01*1.01=1.0201 resistor factors.
ENhigh>=(3.2-0.1-1.2uA*1k*b)/(1+1k*b/(68k*a))
=3.052060850V >1.2V. Includes allocated1uA Q3 off leakage.
U16 high load<=4.6/(68k*a)+1.2uA=70.220568uA <100uA.
ENlow<=0.1+0.2uA*1k*b=0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 Ioff: not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*a)+1uA)/(1/(1k*a)+1/0.2ohm)
=0.938689mV <0.4V; ignores helpful R69.
TI0.2uA EN leakage applies only under its powered test conditions.
Retain1uA adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Ron are acceptance allocations.

Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*4.6/(1k*a)=18.773595mA (not continuous rating).
Isteady<=4*(4.6/(1M*a)+1uA)=22.773595uA.
VGS>=(2.7-0.1-1uA*1k*b)/(1+1k*b/(1M*a))=2.596278V.
SENS-010 held control remainder=1825-1100-500-4-22.773595
=198.226405uA, to cover all other held-island loads.
1100uA reservoir leakage:500uA non-rail inverter-input allocation.
Retain1uC transition charge and10ms complete response allocation.
R34=110k: POR sink=(1.3-0.2)/(110k*a)+3.8uA=14.003041uA.
Keep RUN, KILL, discharge and future USB-clear drains separate.
Source protection and reset redesign remain open; no power-up approval.
Sources: ../design/main_regulator_tps63806.md; SENS-010 revision.

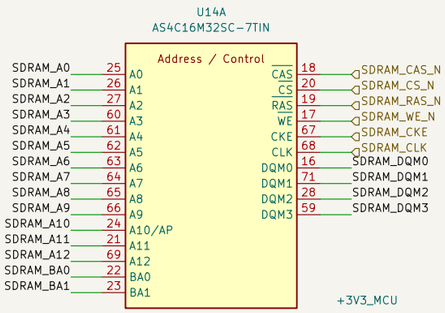
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File: main_3v3_supply.kicad_sch

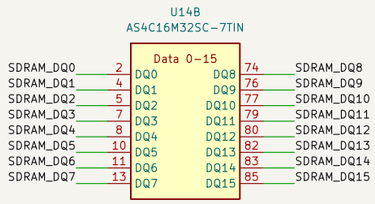
Title: RA8P1 E-reader - main digital 3.3 V supply

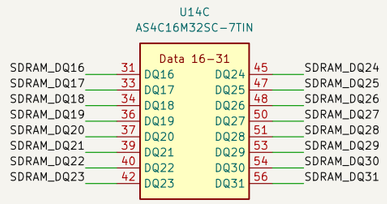
Size: A3	Date: 2026-09-08	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 9/15

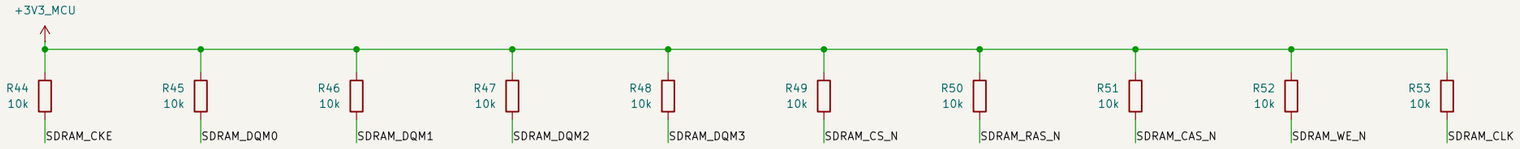
64 MiB SDRAM | Alliance AS4C16M32SC-7TIN
CMS-010: sourcing replacement; public datasheet is PRELIMINARY.
HOLD: released specifications, startup, SI/timing, power and footprint qualification.
See design/component_audit_2026-10-05.md.

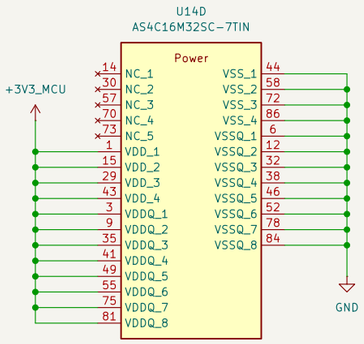
SDRAM_A[0..12]
SDRAM_BA[0..1]
SDRAM_DQM[0..3]
SDRAM_DQ[0..31]

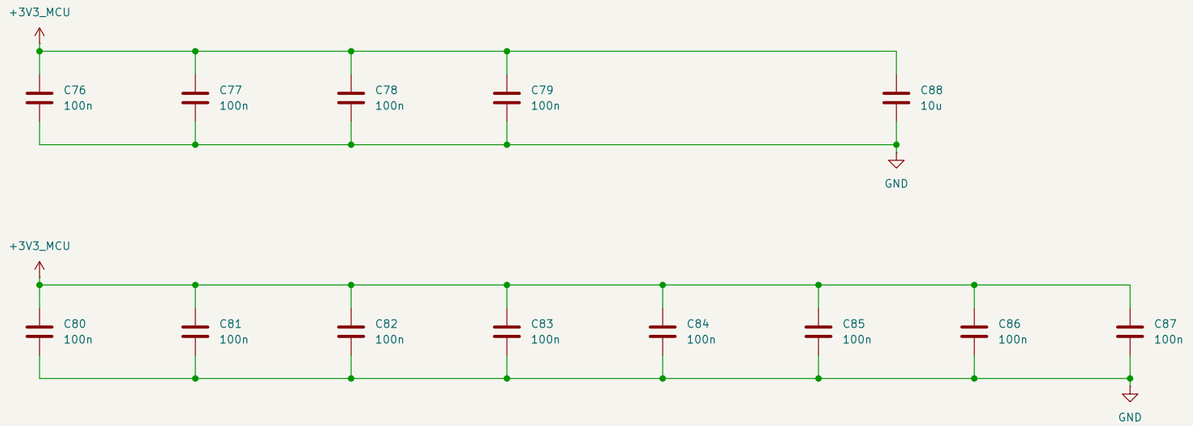












CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

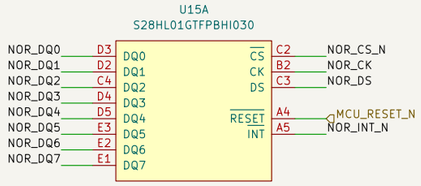
CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V; +/-1%, 100ppm/C, |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

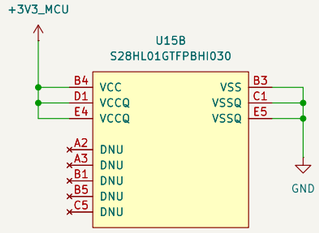
CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail;
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

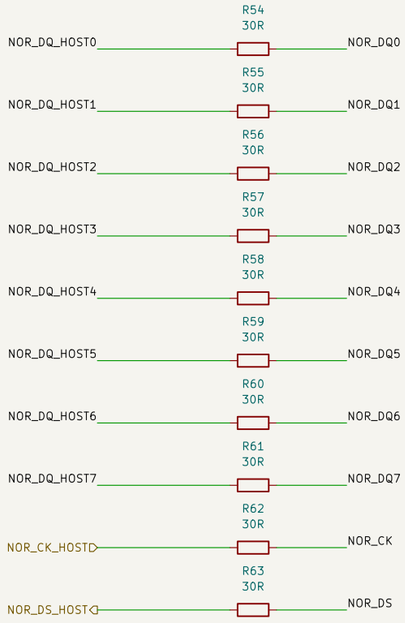
Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

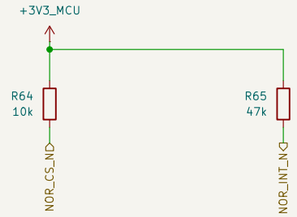
Title: RA8P1 E-reader - 64 MiB SDRAM

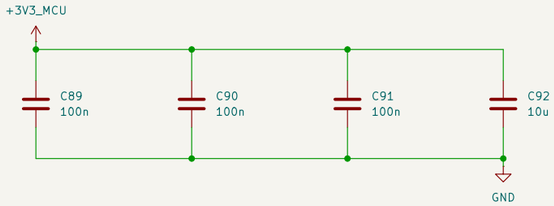
Size: A3	Date: 2026-09-08	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 10/15











CMS-011 | NOR series and idle states
R54-R63: 10 x 30R, 0.1%, 25 ppm/C.
100 C departure from reference temperature:
Rmin = 30 x 0.999 x 0.9975 = 29.895075 ohm
Rmax = 30 x 1.001 x 1.0025 = 30.105075 ohm
Within JESD251 30R +/-1%: 29.7..30.3 ohm.
External parts remain required: SI check OPEN.

R64 CS pull: 10k; R65 INT pull: 47k.
1%, 100 ppm/C, 100 C: factors 0.9801..1.0201.
At 4uA adverse leakage, rail min 3.242044 V:
CS Vhigh >= 3.242044 - 4u x 10201 = 3.201240 V
INT Vhigh >= 3.242044 - 4u x 47944.7 = 3.050265 V
INT Ilow <= 3.6/46064.7 + 2u = 80.150949 uA
< 100uA flash VOL test: 47k departs from 5-10k.
RC timing and board leakage qualification OPEN.

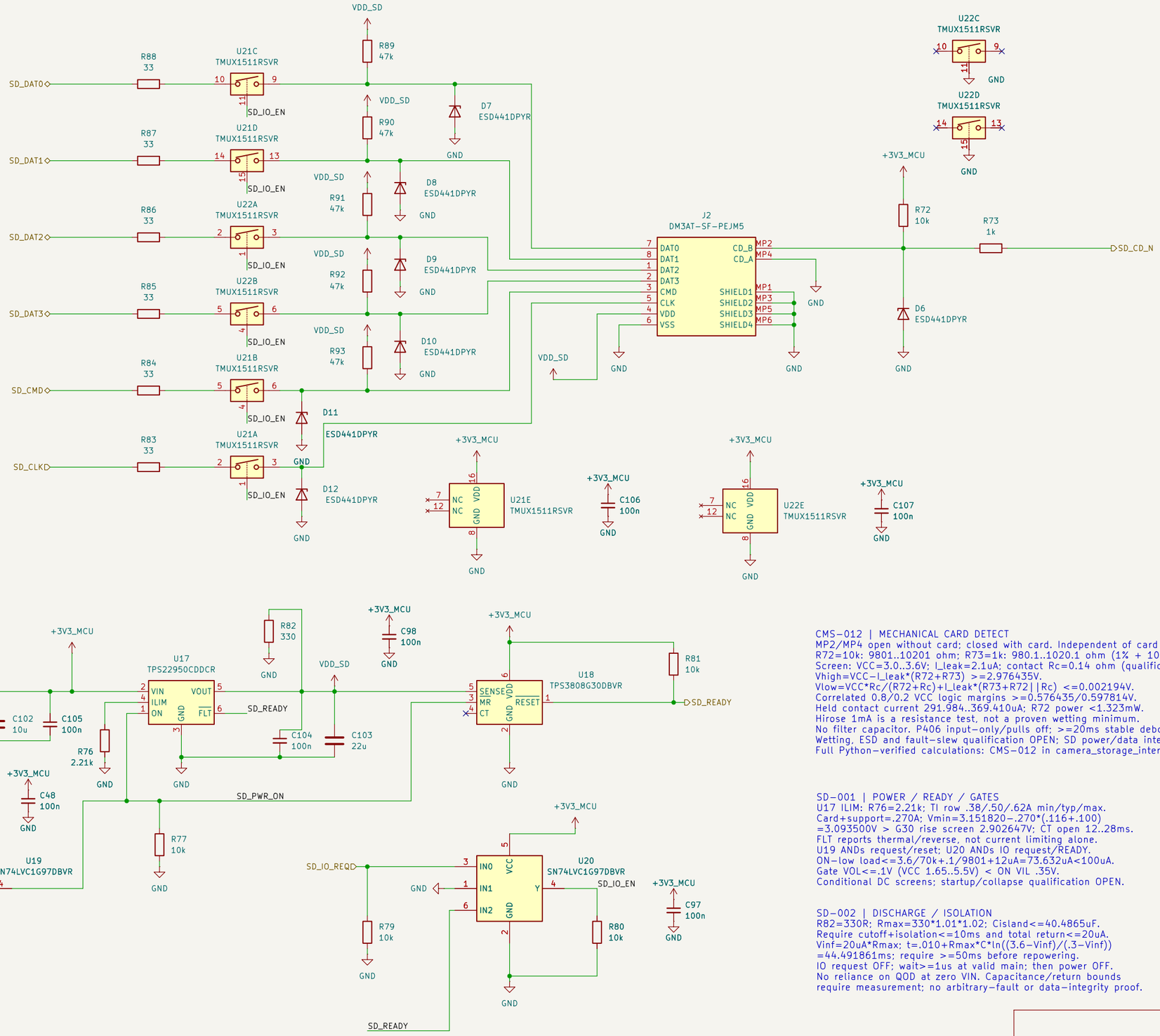
Linked CMS-011B/C/F: sources and Python checks.
MCU hierarchy and common reset connected.
DS is read-only: P801 input, WRMSKMD=0.
CMS-011D defines reset and protocol recovery.

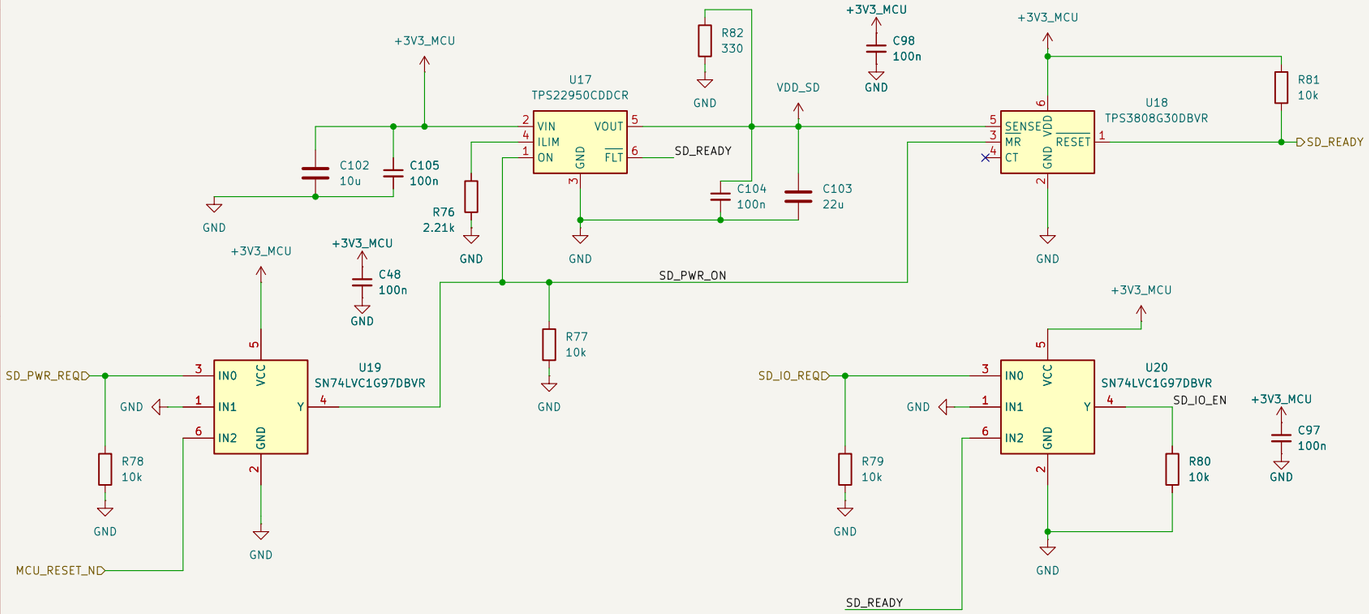
CMS-011 | NOR supply and bypass
U15B VCC and VCCQ share +3V3_MCU (2.7 to 3.6 V).
C89-C91: 3 x 100 nF, one per supply ball.
C92: 10 uF local bulk.
Cnom = 3 x 0.1 + 10 = 10.3 uF.
Initial +/-10%: Cmin/max = 10.3 x (0.9 / 1.1)
= 9.27 / 11.33 uF: not an effective-C guarantee.
Included in the PWR-003 1 mF whole-rail limit.
DC bias, temperature, aging and PDN qualification OPEN.
Linked CMS-011B/F: sources and Python-verified arithmetic.

Sheet: /128 MiB Octal NOR/
File: nor_flash.kicad_sch

Title: RA8P1 E-reader - 128 MiB Octal NOR

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CMS-012 | MECHANICAL CARD DETECT
MP2/MP4 open without card; closed with card. Independent of card VDD.
R72=10k: 9801..10201 ohm; R73=1k: 980.1..1020.1 ohm (1% + 100C TCR).
Screen: VCC=3.0..3.6V; I_leak=2.1uA; contact Rc=0.14 ohm (qualification assumption).
Vhigh=VCC-I_leak*(R72+R73) >=2.976435V.
Vlow=VCC*Rc/(R72+Rc)+I_leak*(R73+R72||Rc) <=0.002194V.
Correlated 0.8/0.2 VCC logic margins >=0.576435/0.597814V.
Held contact current 291.984..369.410uA; R72 power <1.323mW.
Hirose 1mA is a resistance test, not a proven wetting minimum.
No filter capacitor. P406 input-only/pulls off: >=20ms stable debounce REQUIRED.
Wetting, ESD and fault-slew qualification OPEN; SD power/data integrated, qualification pending.
Full Python-verified calculations: CMS-012 in camera_storage_interfaces.md.

SD-001 | POWER / READY / GATES
U17 ILIM: R76=2.21k; TI row .38/.50/.62A min/typ/max.
Card+support=.270A; Vmin=3.151820-.270*(.116+.100)
=3.093500V > G30 rise screen 2.902647V; CT open 12..28ms.
FLT reports thermal/reverse, not current limiting alone.
U19 ANDs request/reset; U20 ANDs IO request/READY.
ON-low load<=3.6/70k+.1/9801+12uA=73.632uA<100uA.
Gate VOL<=.1V (VCC 1.65..5.5V) < ON VIL .35V.
Conditional DC screens; startup/collapse qualification OPEN.

SD-002 | DISCHARGE / ISOLATION
R82=330R; Rmax=330*1.01*1.02; Cisland<=40.4865uF.
Require cutoff+isolation<=10ms and total return<=20uA.
Vinf=20uA*Rmax; t=.010+Rmax*C*ln((3.6-Vinf)/(.3-Vinf))
=44.491861ms; require >=50ms before repowering.
IO request OFF; wait>=1us at valid main; then power OFF.
No reliance on QOD at zero VIN. Capacitance/return bounds
require measurement; no arbitrary-fault or data-integrity proof.

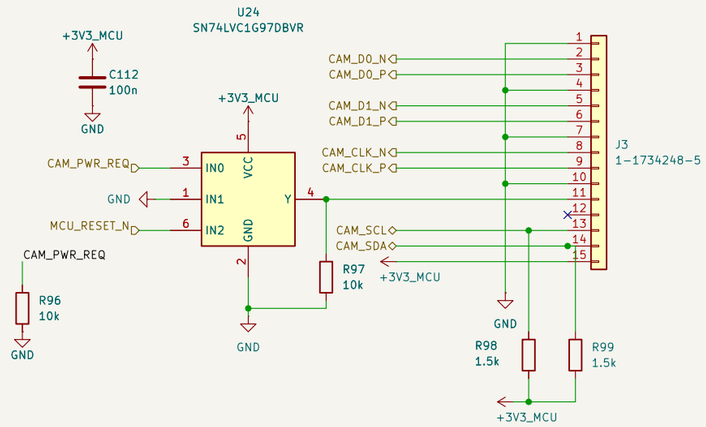
SD-003 | PRE-MIPI SUBTOTAL / RESET FANOUT
Main=.750+.505+.250+.250+.050+.250+.020=2.075A before CMS-013.
CMS-013 adds 20mA: 2.095A (2.090A only with radio 5mA overlap).
Pre-MIPI at Vin 3.2V / assumed 75% efficiency / Vout 3.393012V:
Iin=2.933542A; converter loss=2.346834W; thermal/source OPEN.
SD adds 10.6uF main and 22.1uF switched, plus card.
RST-002: U19 adds 5uA; devices 17uA+overhead 13uA=30uA.
Reset sink=.376190mA<.4mA; high>=2.845789V.
Allocation/conditional math only; firmware and bench acceptance OPEN.

Incomplete electrical design — not for manufacture

Sheet: /microSD removable storage/
File: microsd.kicad_sch

Title: RA8P1 e-reader — microSD removable storage

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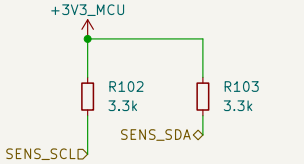
CMS-014 / CMS-015: Pcam 5C, 2-lane CSI-2 to VIN0
U24: PWUP = CAM_PWR_REQ AND MCU_RESET_N.
PWUP low >=100ms; then high >=50ms before SCCB.
Qualification and second-camera/lighting work remain open.
See ../design/camera_storage_interfaces.md, CMS-014..016.

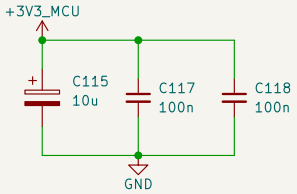
Sheet: /Pcam 5C camera/
File: camera.kicad_sch

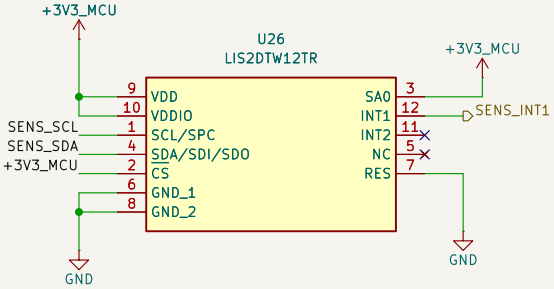
Title:

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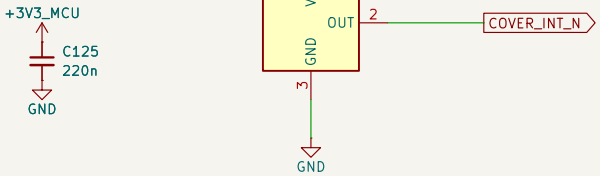
ORIENTATION SENSOR — WIP
U26: LIS2DTW12TR; ST DS12825 Rev 4 Table 1 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE-FN1C100R; place near VDD pin 9 with C117.
R102/R103: 3.3k shared SCL/SDA pull-ups; see ALS-005 bus calculation.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
ST example waits 20ms then polls reset; startup/recovery qualification pending.
CT qty1 sourcing recorded; pin modes, full timing and passive qualification open.
See design/sensor_alternative.md; circuit qualification remains open.



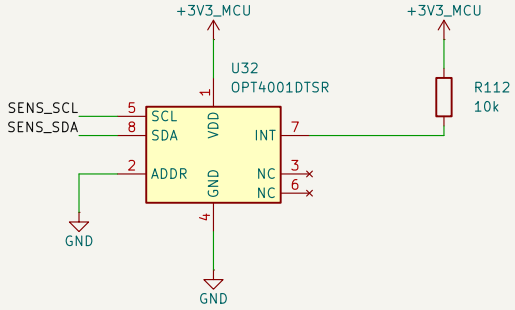




MAGNETIC COVER — WIP
U33 DRV5032FADBZR: FA 20Hz omnipolar push-pull.
VCC1 +3V3_MCU; GND3; OUT2 COVER_INT_N.
MCU P412/M12 IRQ20-DS; input pulls disabled.
C125 220n: local bypass; >=100n effective pending.
LOW = magnet detected; startup delay pending.
Retain +3V3_MCU for wake; rail-off wake unavailable.
Magnet gaps, leakage, power and footprint pending.
See design/hall_cover.md; qualification remains open.





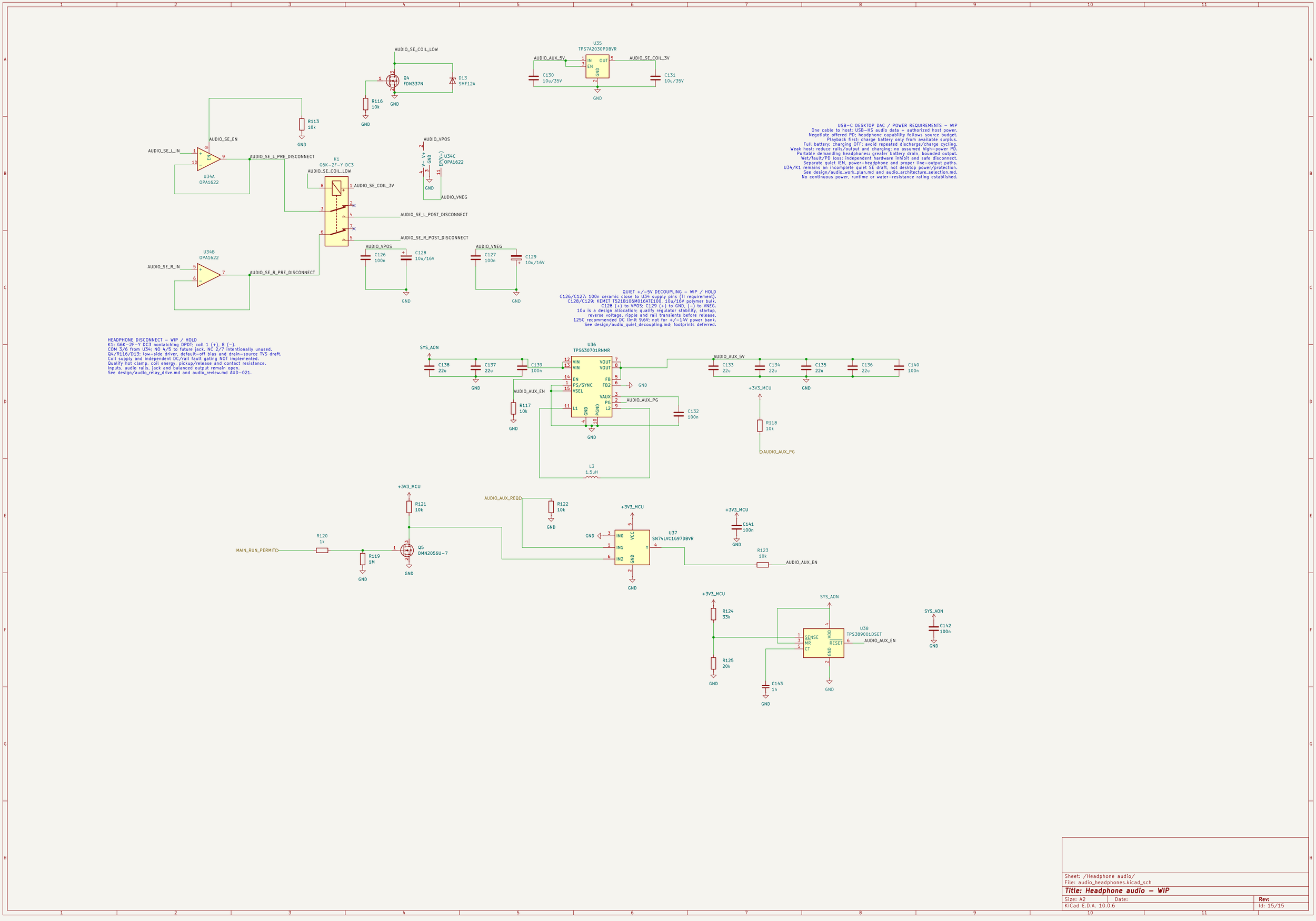


AMBIENT LIGHT — WIP
U32 OPT4001DTSR: TI SBOS993A Rev A, 8-pin DTS.
ADDR grounded: 0x44; C124 local 100nF at VDD pin 1.
R112 10k INT pull-up; firmware keeps INT_DIR=1; polling only.
Shared IIC0_A: R102/R103 3.3k; 25..70pF, +/-17uA planning screen.
Conditional rise 62.912..219.056ns; actual bus and host timing unqualified.
See design/ambient_light.md; sealed optical window/light isolation pending.

Sheet: /Orientation sensor/
File: orientation_sensor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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Sheet: /Headphone audio/
File: audio_headphones.kicad_sch

Title: Headphone audio — WIP

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